Trig Graphs — Practice

For each function below: (i) state the period, (ii) state the phase shift, (iii) state the vertical shift, (iv) determine the domain, (v) determine the range, (vi) list vertical asymptotes in one period, and (vii) sketch one period showing asymptotes.

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1. $f(x) = -2\csc(4x) - 3$
   - Period:
   - Phase shift:
   - Vertical shift:
   - Domain:
   - Range:
   - Asymptotes:
2. $g(x) = 3\sec\left(\frac{1}{2}x + \pi\right) + 1$
   - Period:
   - Phase shift:
   - Vertical shift:
   - Domain:
   - Range:
   - Asymptotes:
3. $h(x) = \frac{1}{2}\tan\left(2x - \frac{\pi}{4}\right) - 2$
   - Period:
   - Phase shift:
   - Vertical shift:
   - Domain:
   - Range:
   - Asymptotes:
4. $k(x) = -\cot(3x) + 4$
   - Period:
   - Phase shift:
   - Vertical shift:
   - Domain:
   - Range:
   - Asymptotes:
5. $m(x) = 2\csc\left(x - \frac{\pi}{6}\right)$
   - Period:
   - Phase shift:
   - Vertical shift:
   - Domain:
   - Range:
   - Asymptotes:
6. $n(x) = -\sec(5x + \pi) + 2$
   - Period:
   - Phase shift:
   - Vertical shift:
   - Domain:
   - Range:
   - Asymptotes:
7. $p(x) = \tan\left(\frac{1}{3}x - \frac{\pi}{2}\right) - 1$
   - Period:
   - Phase shift:
   - Vertical shift:
   - Domain:
   - Range:
   - Asymptotes:
8. $q(x) = -3\cot\left(4x + \frac{\pi}{3}\right) + 1$
   - Period:
   - Phase shift:
   - Vertical shift:
   - Domain:
   - Range:
   - Asymptotes:

Part 1

Solve each equation.

1. $6 = \frac{a}{4} + 2$
2. $9x - 7 = -7$
3. $-4 = \frac{r}{20} - 5$
4. $-6 + \frac{x}{4} = -5$
5. $0 = 4 + \frac{n}{5}$
6. $-1 = \frac{5 + x}{6}$
7. $\frac{v + 9}{3} = 8$
8. $-9x + 1 = -80$
9. $-2 = 2 + \frac{v}{4}$
10. $2(n + 5) = -2$
11. $-6 = \frac{n}{2} - 10$
12. $144 = -12(x + 5)$
13. $8n + 7 = 31$
14. $\frac{n + 5}{-16} = -1$
15. $-10 = 10(k - 9)$
16. $-9x - 13 = -103$
17. $-10 = -10 + 7m$
18. $\frac{m}{9} - 1 = -2$
19. $-243 = -9(10 + x)$
20. $9 + 9n = 9$
21. $7(9 + k) = 84$
22. $8 + \frac{b}{-4} = 5$

Part 2

Solve each equation.

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1. $\frac{r}{10} + 4 = 5$
2. $\frac{n}{2} + 5 = 3$
3. $3p - 2 = -29$
4. $1 - r = -5$
5. $\frac{k - 10}{2} = -7$
6. $\frac{n - 5}{2} = 5$
7. $-9 + \frac{n}{4} = -7$
8. $\frac{9 + m}{3} = 2$
9. $\frac{-5 + x}{22} = -1$
10. $4n - 9 = -9$
11. $\frac{x + 9}{2} = 3$
12. $\frac{-12 + x}{11} = -3$
13. $\frac{-4 + x}{2} = 6$
14. $-5 + \frac{n}{3} = 0$
15. $\frac{p}{4} + 8 = 7$
16. $9 + \frac{n}{4} = 15$
17. $6 + \frac{x}{2} = 4$
18. $\frac{b + 11}{3} = -2$
19. $\frac{a - 10}{3} = -4$
20. $-12r + 4 = 100$
21. $\frac{m}{16} - 9 = -8$
22. $-7 + 4r = -15$
23. $\frac{m - 13}{2} = -8$
24. $-5x + 13 = -17$
25. $\frac{k + 10}{-2} = 5$
26. $\frac{p + 8}{-2} = 10$
27. $-14r - 19 = 303$
28. $\frac{x}{-4} - 5 = -8$

Part 3

Solve each equation.

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1. $5x + 3 = 23$
2. $\dfrac{x-4}{5} = 2$
3. $-3(2x-5) = 9$
4. $7 - 2x = -9$
5. $\dfrac{3x+1}{4} = 5$
6. $0.5x - 2 = 3$
7. $4(x+2) = 3x + 14$
8. $-\frac{2}{3}x = 8$
9. $\dfrac{2x-3}{6} = -1$
10. $9 = \dfrac{x}{-3} + 4$
11. $2(3x-4) = 16$
12. $\dfrac{x}{5} + 7 = 10$
13. $-4 + \dfrac{x}{2} = 6$
14. $5 - 2x = 11$
15. $3(x-4) + 2x = 18$
16. $\dfrac{x+3}{-4} = 2$
17. $0 = -5 + 2n$
18. $\dfrac{6x+9}{3} = 7$
19. $-7 = 2(-3+x)$
20. $\dfrac{4x}{7} = 12$
21. $-3(x+5) = 12$
22. $\dfrac{x}{4} - 3 = -8$
23. $2x + 1 = -3x + 11$
24. $\dfrac{7-x}{2} = 5$

Part 4

Solve each equation or system.

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1. $0x = 5$
2. $0x = 0$
3. $2(x+1) = 2x + 3$
4. $3(x-2) = 3x - 6$
5. $\begin{cases} x + y = 2 \\ 2x + 2y = 5 \end{cases}$
6. $\begin{cases} x + y = 2 \\ 2x + 2y = 4 \end{cases}$
7. $\begin{cases} x + 2y = 3 \\ 2x + 4y = 6 \end{cases}$
8. $\begin{cases} x + 2y = 3 \\ 2x + 4y = 7 \end{cases}$

Part 5

Solve each system of equations.

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1. $\begin{cases} x + y = 5 \\ x - y = 1 \end{cases}$

2. $\begin{cases} 2x + 3y = 7 \\ 4x - y = 1 \end{cases}$

3. $\begin{cases} 3x - 2y = 4 \\ 6x - 4y = 8 \end{cases}$

4. $\begin{cases} 3x - 2y = 4 \\ 6x - 4y = 9 \end{cases}$

5. $\begin{cases} 5x + y = 11 \\ -x + 2y = 1 \end{cases}$

6. $\begin{cases} x + y = 2 \\ x + y = 3 \end{cases}$

7. $\begin{cases} a + 2b = 4 \\ 3a + 6b = 12 \end{cases}$

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§ Groups A **group** is a nonempty set G under the (binary) operation $*: S \times S \to S$, denoted by $(G, *)$, such that

- **Closure:** For all $a, b \in G$, $a * b \in G$.
- **Associativity:** For all $a, b, c \in G$, $(a * b) * c = a * (b * c)$.
- **Existence of Identity:** There exists an element $e \in G$ such that for all $a \in G$, $e * a = a * e = a$.
- **Existence of Inverse:** For each $a \in G$, there exists an element $a^{-1} \in G$ such that $a * a^{-1} = a^{-1} * a = e$.

1. Show that $(\mathbb{Z}, +)$ is a group.

   Proof: Let $G = \mathbb{Z}$ and let the operation be $+$. We will show that $(G, +)$ satisfies all four group properties.

   - **Closure:** For any $a, b \in G$, $a + b \in G$ since the sum of two integers is an integer.
   - **Associativity:** For any $a, b, c \in G$, $(a + b) + c = a + (b + c)$ by the associativity of integer addition.
   - **Existence of Identity:** The identity element is 0 since for any $a \in G$, $0 + a = a + 0 = a$.
   - **Existence of Inverse:** For each $a \in G$, the inverse element is $-a$ since $a + (-a) = (-a) + a = 0$.

   Since all four group properties are satisfied, it follows that $(\mathbb{Z}, +)$ is a group.

   ■

2. Show that $(\mathbb{Z}, \cdot)$ is not a group.

   Proof: Let $G = \mathbb{Z}$ and let the operation be $\cdot$. We will show that $(G, \cdot)$ does not satisfy all four group properties.

   - **Closure:** For any $a, b \in G$, $a \cdot b \in G$ since the product of two integers is an integer.
   - **Associativity:** For any $a, b, c \in G$, $(a \cdot b) \cdot c = a \cdot (b \cdot c)$ by the associativity of integer multiplication.
   - **Existence of Identity:** The identity element is 1 since for any $a \in G$, $1 \cdot a = a \cdot 1 = a$.
   - **Existence of Inverse:** For each $a \in G$, there does not exist an inverse element $a^{-1} \in G$ such that $a \cdot a^{-1} = a^{-1} \cdot a = 1$ unless $a = 1$ or $a = -1$. For example, if $a = 2$, then there is no integer b such that $2 \cdot b = 1$.

   Since the existence of inverse property is not satisfied for all elements in G, it follows that $(\mathbb{Z}, \cdot)$ is not a group.

   ■

3. Show that $(\mathbb{Q}, +)$ is a group.

   Proof: Let $G = \mathbb{Q}$ and let the operation be $+$. We will show that $(G, +)$ satisfies all four group properties.

   - **Closure:** For any $a, b \in G$, $a + b \in G$ since the sum of two rational numbers is a rational number.
   - **Associativity:** For any $a, b, c \in G$, $(a + b) + c = a + (b + c)$ by the associativity of rational number addition.
   - **Existence of Identity:** The identity element is 0 since for any $a \in G$, $0 + a = a + 0 = a$.
   - **Existence of Inverse:** For each $a \in G$, the inverse element is $-a$ since $a + (-a) = (-a) + a = 0$.

   Since all four group properties are satisfied, it follows that $(\mathbb{Q}, +)$ is a group.

   ■

4. Show that $(\mathbb{Q}, \cdot)$ is not a group.

   Proof: Let $G = \mathbb{Q}$ and let the operation be $\cdot$. We will show that $(G, \cdot)$ does not satisfy all four group properties.

   - **Closure:** For any $a, b \in G$, $a \cdot b \in G$ since the product of two rational numbers is a rational number.
   - **Associativity:** For any $a, b, c \in G$, $(a \cdot b) \cdot c = a \cdot (b \cdot c)$ by the associativity of rational number multiplication.
   - **Existence of Identity:** The identity element is 1 since for any $a \in G$, $1 \cdot a = a \cdot 1 = a$.

- **Existence of Inverse:** For each $a \in G$, there does not exist an inverse element $a^{-1} \in G$ such that $a \cdot a^{-1} = a^{-1} \cdot a = 1$ if $a = 0$. For example, if $a = 0$, then there is no rational number b such that $0 \cdot b = 1$.

Since the existence of inverse property is not satisfied for all elements in G, it follows that $(\mathbb{Q}, \cdot)$ is not a group.

■

5. Show that $(\mathbb{Q}^*, \cdot)$ is a group, where $\mathbb{Q}^* = \{q \in \mathbb{Q} \mid q \neq 0\}$.

Proof: Let $G = \mathbb{Q}^*$ and let the operation be $\cdot$. We will show that $(G, \cdot)$ satisfies all four group properties.

- **Closure:** For any $a, b \in G$, $a \cdot b \in G$ since the product of two nonzero rational numbers is a nonzero rational number.
- **Associativity:** For any $a, b, c \in G$, $(a \cdot b) \cdot c = a \cdot (b \cdot c)$ by the associativity of rational number multiplication.
- **Existence of Identity:** The identity element is 1 since for any $a \in G$, $1 \cdot a = a \cdot 1 = a$.
- **Existence of Inverse:** For each $a \in G$, the inverse element is $a^{-1} = \frac{1}{a}$ since $a \cdot a^{-1} = a^{-1} \cdot a = 1$.

Since all four group properties are satisfied, it follows that $(\mathbb{Q}^*, \cdot)$ is a group.

■

6. Show that $(\mathbb{R}^+, +)$ is a group.

Proof: Let $G = \mathbb{R}^+$ and let the operation be $+$. We will show that $(G, +)$ does not satisfy all four group properties.

- **Closure:** For any $a, b \in G$, $a + b \in G$ since the sum of two positive real numbers is a positive real number.
- **Associativity:** For any $a, b, c \in G$, $(a + b) + c = a + (b + c)$ by the associativity of real number addition.
- **Existence of Identity:** There does not exist an identity element $e \in G$ such that for all $a \in G$, $e + a = a + e = a$ since the only candidate for the identity element is 0, which is not in G.
- **Existence of Inverse:** For each $a \in G$, there does not exist an inverse element $a^{-1} \in G$ such that $a + a^{-1} = a^{-1} + a = e$ since the only candidate for the inverse element is $-a$, which is not in G.

Since the existence of identity and existence of inverse properties are not satisfied, it follows that $(\mathbb{R}^+, +)$ is not a group.

■

7. Show that $(\mathbb{R}^+, \cdot)$ is a group.

Proof: Let $G = \mathbb{R}^+$ and let the operation be $\cdot$. We will show that $(G, \cdot)$ satisfies all four group properties.

- **Closure:** For any $a, b \in G$, $a \cdot b \in G$ since the product of two positive real numbers is a positive real number.
- **Associativity:** For any $a, b, c \in G$, $(a \cdot b) \cdot c = a \cdot (b \cdot c)$ by the associativity of real number multiplication.
- **Existence of Identity:** The identity element is 1 since for any $a \in G$, $1 \cdot a = a \cdot 1 = a$.
- **Existence of Inverse:** For each $a \in G$, the inverse element is $a^{-1} = \frac{1}{a}$ since $a \cdot a^{-1} = a^{-1} \cdot a = 1$.

Since all four group properties are satisfied, it follows that $(\mathbb{R}^+, \cdot)$ is a group.

■

§ Abelian Groups A group $(G, *)$ is called an **Abelian group** (or **commutative group**) if (and only if) **commutative**, that is,

$$\text{For all } a, b \in G,\ a * b = b * a.$$

1. Let G be a nonempty set of 2×2 matrices. Then, $(G, \cdot)$ is not an Abelian group, where $\cdot$ is the matrix multiplication operation.

Proof: Let G be the set of all 2×2 matrices with real entries, and let the operation be matrix multiplication $\cdot$. We will show that $(G, \cdot)$ does not satisfy the commutative property. Consider the following two matrices in G:

$$A = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 0 \\ 3 & 1 \end{pmatrix}.$$

Now, we compute $A \cdot B$ and $B \cdot A$:

$$A \cdot B = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} 1 & 0 \\ 3 & 1 \end{pmatrix} = \begin{pmatrix} 1+6 & 2+2 \\ 0+3 & 0+1 \end{pmatrix} = \begin{pmatrix} 7 & 4 \\ 3 & 1 \end{pmatrix},$$

and

$$B \cdot A = \begin{pmatrix} 1 & 0 \\ 3 & 1 \end{pmatrix} \cdot \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1+0 & 2+0 \\ 3+0 & 6+1 \end{pmatrix} = \begin{pmatrix} 1 & 2 \\ 3 & 7 \end{pmatrix}.$$

Since $A \cdot B = \begin{pmatrix} 7 & 4 \\ 3 & 1 \end{pmatrix}$ is not equal to $B \cdot A = \begin{pmatrix} 1 & 2 \\ 3 & 7 \end{pmatrix}$. Therefore, $(G, \cdot)$ does not satisfy the commutative property, and hence it is not an Abelian group.

■

§ Finite Groups A group $(G, *)$ is called a **finite group** if (and only if) the order of G, denoted by $|G|$, is finite; that is,

$$|G| < \infty.$$

§ Group A nonempty set G with a binary operation $* : G \times G \to G$, written $(G, *)$, is a **group** if and only if

- **Closure:** For all $a, b \in G$, $a * b \in G$.
- **Associativity:** For all $a, b, c \in G$, $(a * b) * c = a * (b * c)$.
- **Identity:** There exists $e \in G$ such that for all $a \in G$, $e * a = a * e = a$.
- **Inverses:** For each $a \in G$ there exists $b \in G$ such that $a * b = b * a = e$.

Groups are generalizations of the familiar products of addition and of multiplication of real numbers. Remember, though, that a set with a product is a group if and only if the four axioms are satisfied!

For instance, $\mathbb{Z}$ is a group under addition but not under multiplication (for more than one reason). The set $\mathbb{Q}$ is also group under addition but not under multiplication. The set $\mathbb{R}^+$of positive real numbers is a group under multiplication but not under addition.

The groups above are infinite, but many interesting groups are finite. The order of a group G is the number of elements in G, denoted $|\boldsymbol{G}|$. If (and only if) $|\boldsymbol{G}| < \infty$, then G is a finite group.

The smallest group consists of a single (identity) element, usually denoted e (or sometimes i when the group elements are functions). Here's an example of a group of order 2: Claim: The set $\{-1, 1\}$ under multiplication is a group. Proof: Consider the set $\{-1, 1\}$ under multiplication, denoted $\cdot$. Since $-1 \cdot -1 = 1 = 1 \cdot 1$ and $-1 \cdot 1 = -1 = 1 \cdot -1$, then $\{-1, 1\}$ is closed under multiplication. Since multiplication of real number is associative, then multiplication on $\{-1, 1\}$ is associative. Since $-1 \cdot 1 = 1 \cdot -1 = -1$ and $1 \cdot 1 = 1$, then 1 is the identity under multiplication. Since $-1 \cdot -1 = 1$ and $1 \cdot 1 = 1$, then both -1 and 1 have multiplicative inverses. Since all four axioms are satisfied, $(\{-1, 1\}, \cdot)$ is a group. QED

When the order of a group is small, it is often very useful to write out the product table for the group.

For instance, let G be a group of order two with product $*$. Then $G = \{e, a\}$ where e is the identity and a denotes the single non-identity element. The product table for G is:

$*$	e	a
e	e	a
a	a	e

This is the "only" group of order 2. By which we mean that all groups of order 2 must be isomorphic to this group.

§ Group Isomorphism Let $(G, *)$ and $(G', \star)$ be groups. An **isomorphism** is a bijection $\phi : G \to G'$ such that $\phi(a * b) = \phi(a) \star \phi(b)$ for all $a, b \in G$. We write $G \simeq G'$ if such a map exists.

Claim: The group $(\{-1, 1\}, \cdot)$ is isomorphic to the group $(G, *)$ with product table above. Proof: Let $\phi : \{-1, 1\} \to \{e, a\}$ be the function given by $\phi(1) = e$ and $\phi(-1) = a$. Then by inspection ϕ is one-to-one and onto, and hence a bijection. Also, $\phi(1 \cdot 1) = \phi(1) = e = e*e = \phi(1)*\phi(1)$, $\phi(1 \cdot -1) = \phi(-1) = a = e*a = \phi(1)*\phi(-1)$, $\phi(-1 \cdot 1) = \phi(-1) = a = a*e = \phi(-1)*\phi(1)$, and $\phi(-1 \cdot -1) = \phi(1) = e = a * a = \phi(-1) * \phi(-1)$. Hence, ϕ preserve operations, and $(\{-1, 1\}, \cdot) \simeq (\{e, a\}, *)$. QED Can you think of another group of order 2 ? What about the set $\{[0]_2, [1]_2\}$ of congrunence classes modulo two under addition? Can you prove this is isomorphic to $(G, *)$ with product table above?

Fill in the product tables below. Let e denote the identity element. Hint: How many times can an element appear in a given row (or column)? How many times must an element appear in a given row (or column)? If $g * h = h$ in a group $(G, *)$ with $g, h \in G$, then what is known about g ?

Up to isomorphism, there is one group of order 3. You should understand why after filling in this product table.

	e	a	b
e			
a			
b			

Up to isomorphism, there are two groups of order 4. Identify the first one by completing this product table.

	e	a	b	c
e				
a		b		
b				
c				

Identify the other group of order 4 which is not isomorphic to the previous one. Hint: confirm your answer is not isomorphic to the previous group of order 4 under bijections like $b \leftrightarrow c$ or $a \leftrightarrow b$.

	e	a	b	c
e				
a				
b				
c				
c				

Recall Wbwk8, Problem 4. Fix $n \geq 2$ and consider $\mathbb{Z}_n$, the set of equivalence classes under the relation congruence modulo n on $\mathbb{Z}$. Are the following propositions true or false?

1. Since addition on $\mathbb{Z}$ is associative, so is addition on $\mathbb{Z}_n$.
2. Since multiplication on $\mathbb{Z}$ is commutative, so is multiplication on $\mathbb{Z}_n$.
3. For all $a, b, c, d \in \mathbb{Z}$, if $[a] = [b]$ and $[c] = [d]$, then $[ac] = [bd]$.
4. Multiplication is well-defined on $\mathbb{Z}_n$.
5. Since multiplication on $\mathbb{Z}$ is associative, so is multiplication on $\mathbb{Z}_n$.
6. Addition is well-defined on $\mathbb{Z}_n$.
7. For all $a, b, c, d \in \mathbb{Z}$, if $[a] = [b]$ and $[c] = [d]$, then $[a + c] = [b + d]$.
8. Since addition on $\mathbb{Z}$ is commutative, so is addition on $\mathbb{Z}_n$.

Now, compute the following sum and product tables for $n = 6$.

$+$	[0]	[1]	[2]	[3]	[4]	[5]
[0]						
[1]						
[2]						
[3]						
[4]						
[5]						

$\cdot$	[0]	[1]	[2]	[3]	[4]	[5]
[0]						
[1]						
[2]						
[3]						
[4]						
[5]						

Fix $n \geq 2$ and consider $\mathbb{Z}_n$, the set of equivalence classes under the relation congruence modulo n on $\mathbb{Z}$. By definition, $\mathbb{Z}_n^* = \mathbb{Z}_n \setminus \{[0]\}$. Are the following propositions true or false?

(**false**) Under multiplication $[0]$ is the identity element in $\mathbb{Z}_n$.

(**false**) The set $\mathbb{Z}_n$ is a group under the product of multiplication.

(**true**) There exists an identity element in $\mathbb{Z}_n$ under addition.

(**false**) The set $\mathbb{Z}_n^*$ is a group under the product of multiplication.

(**false**) For every $[a] \in \mathbb{Z}_n$ there exists a $[b] \in \mathbb{Z}_n$ such that $[ab]$ is the multiplicative identity.

(**true**) Under multiplication $[1]$ is the identity element in $\mathbb{Z}_n^*$.

(**false**) Under multiplication $[0]$ is the identity element in $\mathbb{Z}_n^*$ when n is prime.

(**true**) Under addition $[0]$ is the identity element in $\mathbb{Z}_n$.

(**true**) There exist inverses in $\mathbb{Z}_n^*$ under multiplication when n is prime.

(**true**) There exists an identity element in $\mathbb{Z}_n^*$ under multiplication.

(**true**) The set $\mathbb{Z}_n^*$ is a group under the product of multiplication when n is prime.

(**false**) For every $[a] \in \mathbb{Z}_n^*$ there exists a $[b] \in \mathbb{Z}_n^*$ such that $[ab]$ is the multiplicative identity.

(**true**) For every $[a] \in \mathbb{Z}_n$ there exists a $[b] \in \mathbb{Z}_n$ such that $[a + b]$ is the additive identity.

(**true**) Under multiplication $[1]$ is the identity element in $\mathbb{Z}_n^*$ when n is prime.

(**true**) There exists an identity element in $\mathbb{Z}_n$ under multiplication.

(**false**) There exist inverses in $\mathbb{Z}_n$ under multiplication.

(**false**) Under multiplication $[0]$ is the identity element in $\mathbb{Z}_n^*$.

(**true**) For every $[a] \in \mathbb{Z}_n^*$ there exists a $[b] \in \mathbb{Z}_n^*$ such that $[ab]$ is the multiplicative identity when n is prime.

(**false**) Under addition $[1]$ is the identity element in $\mathbb{Z}_n$.

(**true**) The set $\mathbb{Z}_n$ is a group under the product of addition.

(**true**) Under multiplication $[1]$ is the identity element in $\mathbb{Z}_n$.

(**true**) There exists an identity element in $\mathbb{Z}_n^*$ under multiplication when n is prime.

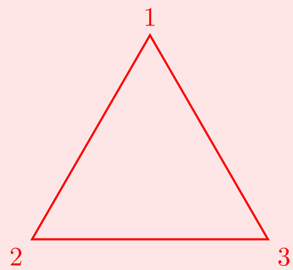
(**true**) There exist inverses in $\mathbb{Z}_n$ under addition.

(**false**) There exist inverses in $\mathbb{Z}_n^*$ under multiplication.

§ Nonabelian Group: Symmetries of an Equilateral Triangle A group $(G, *)$ is **abelian** if and only if $a * b = b * a$ for all $a, b \in G$. While many familiar groups are abelian, most groups are not! Hence, it is very important to understand well some examples of nonabelian groups.

The smallest nonabelian group has order 6. One way to understand this group is as the symmetries of an equilateral triangle.

Consider an equilateral triangle with vertex labels 1, 2, and 3:



Let f be a reflection through the vertical axis. So f fixes point 1 and switches 2 and 3.

Let g be a rotation 120° clockwise around the center point. So g sends 1 to 2, 2 to 3, and 3 to 1.

There are four other symmetries of an equilateral triangle. What are they in terms of either an axis of reflection or an angle of rotation?

The composition of two symmetries is another symmetry, and this product is associative. There is an identity symmetry, and every symmetry has an inverse. Why? How would you justify these statements?

Hence, the symmetries of an equilateral triangle form a group under composition.

Match each symmetry on the left with an equivalent one from the right if possible. The notation fg means $f \circ g$, so first rotate 120°, then reflect through the vertical axis.

A. gf

B. g^{-1}

C. $g^2 f$

D. f

E. f^2

F. None of the above

Match the following.

1. f^{-1}
2. g^3
3. fg^2
4. fg
5. g^2

Fill in the product table below using the given labels for the symmetries. Let e denote the identity symmetry. Enter g^2 as "gg" and fg^2 as "fgg".

	e	f	g	g^2	fg	fg^2
e						
f						
g						
g^2						
fg						
fg^2						

§ Order of an Element Let $(G, *)$ be a group with identity e. The **order** of an element $a \in G$, denoted $o(a)$, is the least positive integer m such that $a^m = e$. If no such m exists, we say a has **infinite order**.

Let $(G, *)$ be the group of symmetries of an equilateral triangle under composition. What are the truth values of the following propositions?

1. Every element in $(G, *)$ has order 2 or 3.
2. $fg = g^{-1}f$ but $g \neq g^{-1}$.
3. The order of $(G, *)$ is 6.
4. The order of g^2 is 6.
5. The order of f is 2.
6. $(fg)^2 \neq f^2g^2$.
7. For all elements $h \in G$, $h^6 = e$.
8. The order of g is 3.
9. The order of every element in $(G, *)$ divides the order of the group.
10. $g^2(fg) = (fg)g^2$.
11. $(G, *)$ is an abelian group.

Up to isomorphism, there is a unique group of order 5 . Let $G = \{a, b, c, d, e\}$ be a group with product *. Complete the product table for $(G, *)$ below.

$*$	e	a	b	c	d
e	e	a	b	c	d
a	a	b	d	e	c
b	b	d	c	a	e
c	c	e	a	d	b
d	d	c	e	b	a

The congruence classes modulo 5 are also a group of order 5 under addition. Show that ($\mathbb{Z}_5, +$) is isomorphic to $(G, *)$ in several different ways by completing the various bijections $\phi : \mathbb{Z}_5 \to G$ below.

Match the elements of $\mathbb{Z}_5$ on the left with their images in G. Remember that the bijection ϕ must preserve operations to be an isomorphism.

1. Suppose ϕ maps [1] to a.

   Answer:

   - $\phi([0]) \to e$
   - $\phi([1]) \to a$
   - $\phi([2]) \to a^2 = b$
   - $\phi([3]) \to a^3 = d$
   - $\phi([4]) \to a^4 = c$

   □

2. Suppose ϕ maps [1] to b.

   Answer:

   - $\phi([0]) \to e$
   - $\phi([1]) \to b$
   - $\phi([2]) \to b^2 = c$
   - $\phi([3]) \to b^3 = a$
   - $\phi([4]) \to b^4 = d$

   □

3. Suppose ϕ maps [1] to c.

   Answer:

   - $\phi([0]) \to e$
   - $\phi([1]) \to c$
   - $\phi([2]) \to c^2 = d$
   - $\phi([3]) \to c^3 = b$
   - $\phi([4]) \to c^4 = a$

   □

4. Suppose ϕ maps [1] to d.

   Answer:

- $\phi([0]) \to e$ (identity: $d^0 = e$)
- $\phi([1]) \to d$ ($d^1 = d$)
- $\phi([2]) \to d^2 = a$ ($d^2 = a$ from the table)
- $\phi([3]) \to d^3 = c$ ($d^3 = c$ from the table)
- $\phi([4]) \to d^4 = b$ ($d^4 = b$ from the table)

$\square$

5. Suppose ϕ maps [2] to a.

Answer:

- $\phi([0]) \to e$ (identity: $a^0 = e$)
- $\phi([1]) \to d$ ($a^1 = a$, but $a = d^2$, so d is the element after e in the cycle)
- $\phi([2]) \to a$ (by assumption)
- $\phi([3]) \to c$ ($a^3 = c$ from the table)
- $\phi([4]) \to b$ ($a^4 = b$ from the table)

$\square$

6. Suppose ϕ maps [3] to a.

Answer:

- $\phi([0]) \to e$ (identity: $a^0 = e$)
- $\phi([1]) \to b$ ($a^1 = a$, $a^2 = b$)
- $\phi([2]) \to c$ ($a^3 = c$)
- $\phi([3]) \to a$ (by assumption)
- $\phi([4]) \to d$ ($a^4 = d$)

$\square$

7. Suppose ϕ maps [4] to a.

Answer:

- $\phi([0]) \to e$
- $\phi([1]) \to c$
- $\phi([2]) \to d$
- $\phi([3]) \to b$
- $\phi([4]) \to a$

$\square$

§ Subgroup A nonempty subset H of G is called a subgroup of G if (and only if) H itself is a group under the product of G.

§ Subgroup Lemma A subset $A \subseteq G$ is a subgroup of $(G, *)$ if and only if

1. A is nonempty,
2. for all $a, b \in A, a * b \in A$, and
3. for all $a \in A, a^{-1} \in A$

Select all subgroups of the group $\mathbb{Z}_5$ under addition.

1. $\{[1]\}$
2. $\{[0], [2], [4]\}$
3. $\{[0], [1], [4]\}$
4. $\{[0], [3]\}$
5. $\{[1], [2], [3]\}$
6. $\{[0], [1], [3]\}$
7. $\{[0], [1]\}$
8. $\{[0], [1], [2], [3], [4]\}$
9. $\{[0]\}$
10. $\{[1], [2], [3], [4]\}$

Answer: I, H (i.e. $\{[0]\}$, $\mathbb{Z}_5$).

□

Select all subgroups of the group $\mathbb{Z}_6$ under addition.

A. $\{[2], [4]\}$
B. $\{[0]\}$
C. $\{[1], [2], [3], [4], [5]\}$
D. $\{[0], [2], [4]\}$
E. $\{[0], [2], [3], [4]\}$
F. $\{[0], [1], [3], [5]\}$
G. $\{[1]\}$
H. $\{[0], [1], [2], [3], [4], [5]\}$
I. $\{[3]\}$
J. $\{[0], [3]\}$

Answer: B, D, J, H (i.e. $\{[0]\}$, $\{[0], [2], [4]\}$, $\{[0], [3]\}$, $\mathbb{Z}_6$).

□

Select all subgroups of the group $\mathbb{Z}_5^*$ under multiplication.

A. $\{[0], [1], [2], [3], [4]\}$
B. $\{[1]\}$

C. $\{[0]\}$

D. $\{[3]\}$

E. $\{[2], [4]\}$

F. $\{[1], [4]\}$

G. $\{[2], [3], [4]\}$

H. $\{[0], [1], [3]\}$

I. $\{[1], [3]\}$

J. $\{[1], [2], [3], [4]\}$

Answer: B, F, J (i.e. $\{[1]\}$, $\{[1], [4]\}$, $\mathbb{Z}_5^*$).

$\square$

As you know from Webwork 10, Problem 1, there are two non-isomorphic groups of order 4. The second one from that problem is named the **Klein 4-group** and denoted V_4.

V_4 has the group presentation $< a, b \mid a^2 = e, b^2 = e, (ab)^2 = e >$ where e is the identity element. Hence, the elements a and b generate the group under the relations $a^2 = e$, $b^2 = e$, and $(ab)^2 = e$.

Match each element on the left with an equivalent one from the right if possible.

Elements to Match:

1. aba
2. a^{-1}
3. $(ab)^{-1}$
4. ba
5. b^{-1}
6. bab

Options:

A. a

B. ab

C. b

D. None of the above

Answer: The given relations imply several key properties:

- $a^2 = e \implies a^{-1} = a$.
- $b^2 = e \implies b^{-1} = b$.
- $(ab)^2 = e \implies (ab)^{-1} = ab$.
- The relations also imply the group is abelian: $(ab)(ab) = e$. Multiplying on the left by a gives $a(abab) = ae \implies (a^2)bab = a \implies bab = a$. Multiplying this on the right by b gives $(bab)b = ab \implies ba(b^2) = ab \implies ba = ab$.

Using these properties, we can simplify each expression:

- 1. $aba = a(ba) = a(ab) = (a^2)b = eb = b \quad \rightarrow$ **C**
- 2. $a^{-1} = a \quad \rightarrow$ **A**
- 3. $(ab)^{-1} = ab \quad \rightarrow$ **B**
- 4. $ba = ab \quad \rightarrow$ **B**
- 5. $b^{-1} = b \quad \rightarrow$ **C**
- 6. $bab = (ba)b = (ab)b = a(b^2) = ae = a \quad \rightarrow$ **A**

$\square$

Fill in the product table below using the given labels for the elements.

$\circ$	e	a	b	ab
e	e	a	b	ab
a	a	e	ab	b
b	b	ab	e	a
ab	ab	b	a	e

Select all subgroups of Klein 4-group.

A. $\{e, b, ab\}$

B. $\{e, a\}$

C. $\{e\}$

D. $\{\}$

E. $\{a, b, ab\}$

F. $\{e, b\}$

G. $\{b, ab\}$

H. $\{a, b\}$

I. $\{e, a, ab\}$

J. $\{a, ab\}$

K. $\{a\}$

L. $\{b\}$

M. $\{e, a, b\}$

N. $\{e, a, b, ab\}$

O. $\{e, ab\}$

P. $\{ab\}$

Answer: To be a subgroup, a subset must contain the identity element e, be closed under the operation, and contain the inverse of each of its elements.

- Options D, E, G, H, J, K, L, P fail because they do not contain the identity e.
- Options A, I, M fail the closure property (e.g., in A, $b \circ ab = a$, which is not in the set).

The correct options are: **B, C, F, N, O**
(i.e., $\{e, a\}$, $\{e\}$, $\{e, b\}$, $\{e, a, b, ab\}$, and $\{e, ab\}$).

□

Problem 1: $(\mathbb{Z}_4, +)$ vs. V_4

To demonstrate that two groups are not isomorphic, one must find a characteristic which is different between the groups. Select all characteristics which could be used to show that V_4 is not isomorphic to $(\mathbb{Z}_4, +)$.

A. $(\mathbb{Z}_4, +)$ has an element of order 4 but V_4 does not.

B. Every element of V_4 is its own inverse, but $[1], [3] \in \mathbb{Z}_4$ with $[1]^{-1} = [3]$ and $[3]^{-1} = [1]$.

C. $(\mathbb{Z}_4, +)$ has an element of order 1 but V_4 does not.

D. V_4 has a subgroup of order 1 but $(\mathbb{Z}_4, +)$ does not.

E. V_4 has an element of order 2 but V_4 does not.

F. $(\mathbb{Z}_4, +)$ is abelian but V_4 is not.

G. $(\mathbb{Z}_4, +)$ has one element of order 2 while V_4 has three.

H. V_4 has a subgroup of order 4 but $(\mathbb{Z}_4, +)$ does not.

I. V_4 is cyclic but $(\mathbb{Z}_4, +)$ is not. See problem 5.

J. V_4 has a subgroup of order 2 but $(\mathbb{Z}_4, +)$ does not.

K. V_4 has three subgroups of order 2 while $(\mathbb{Z}_4, +)$ only has one.

L. $(\mathbb{Z}_4, +)$ is cyclic but V_4 is not. See problem 5.

M. V_4 is abelian but $(\mathbb{Z}_4, +)$ is not.

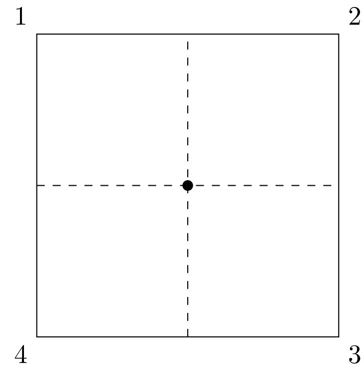
N. None of the above

Answer: The correct statements that distinguish the two groups are **A, B, G, K, L**.

- **A**: True. $(\mathbb{Z}_4, +)$ has elements $[1]$ and $[3]$ of order 4. All non-identity elements in V_4 have order 2.
- **B**: True. This is a restatement of the order property. In V_4, $x^2 = e$ for all x, so $x = x^{-1}$. In $\mathbb{Z}_4$, $[1]^{-1} = [3]$.
- **C**: False. Both groups have an identity element of order 1.
- **D**: False. Both groups have a trivial subgroup $\{e\}$ (or $\{[0]\}$) of order 1.
- **E**: False. Both groups have elements of order 2.
- **F**: False. Both groups are abelian.
- **G**: True. $(\mathbb{Z}_4, +)$ has one element of order 2, $[2]$. V_4 has three elements of order 2, a, b, and ab.
- **H**: False. Both groups have a subgroup of order 4 (the group itself).
- **I**: False. $(\mathbb{Z}_4, +)$ is cyclic, V_4 is not.
- **J**: False. Both groups have subgroups of order 2.
- **K**: True. V_4 has three subgroups of order 2 ($\{e, a\}$, $\{e, b\}$, $\{e, ab\}$). $(\mathbb{Z}_4, +)$ has one ($\{[0], [2]\}$).
- **L**: True. $(\mathbb{Z}_4, +)$ is cyclic (generated by $[1]$). V_4 is not cyclic.
- **M**: False. Both groups are abelian.

$\square$

The Klein 4-group is (isomorphic to) the group of symmetries of a non-square rectangle:



Let x be a reflection through the vertical axis, y a reflection through the horizontal axis, z a rotation 180 degrees around the midpoint, and i the identity symmetry. Let $G = \{x, y, z, i\}$ be the group of symmetries under composition.

Which of the following is **not** an isomorphism from V_4 to G?

A. $\phi(e) = x, \phi(a) = i, \phi(b) = y, \phi(ab) = z$.

B. $\phi(e) = z, \phi(a) = x, \phi(b) = i, \phi(ab) = y$.

C. $\phi(e) = i, \phi(a) = y, \phi(b) = z, \phi(ab) = x$.

D. $\phi(e) = i, \phi(a) = z, \phi(b) = y, \phi(ab) = i$.

E. $\phi(e) = i, \phi(a) = z, \phi(b) = z, \phi(ab) = y$.

F. $\phi(e) = i, \phi(e) = y, \phi(a) = y, \phi(b) = x, \phi(ab) = z$.

G. $\phi(e) = i, \phi(a) = z, \phi(b) = x, \phi(ab) = y$.

H. $\phi(e) = i, \phi(a) = x, \phi(b) = z, \phi(ab) = y$.

I. $\phi(e) = i, \phi(a) = y, \phi(b) = x, \phi(ab) = z$.

J. $\phi(e) = i, \phi(a) = z, \phi(b) = y, \phi(ab) = x$.

K. $\phi(e) = i, \phi(a) = y, \phi(b) = x, \phi(ab) = x$.

L. $\phi(e) = i, \phi(a) = y, \phi(b) = x, \phi(ab) = z$.

M. None of the above

Answer: A mapping $\phi : V_4 \to G$ is **not** an isomorphism if it fails to be a bijection or fails to map the identity element e to the identity element i.

- **A**: Fails. Does not map the identity correctly ($\phi(e) \neq i$).
- **B**: Fails. Does not map the identity correctly ($\phi(e) \neq i$).
- **C**: This is a valid isomorphism.
- **D**: Fails. Not a bijection (not one-to-one: $\phi(e) = \phi(ab) = i$).
- **E**: Fails. Not a bijection (not one-to-one: $\phi(a) = \phi(b) = z$).
- **F**: Fails. Not a valid function (maps e to two values).
- **G**: This is a valid isomorphism.
- **H**: This is a valid isomorphism.
- **I**: This is a valid isomorphism.

- **J**: This is a valid isomorphism.
- **K**: Fails. Not a bijection (not one-to-one: $\phi(b) = \phi(ab) = x$).
- **L**: This is a valid isomorphism (identical to I).

The correct selections are **A, B, D, E, F, and K**.

$\square$

The symmetric group of degree n, denoted S_n, is the group of one-to-one mappings of a finite set of n elements onto itself. The elements of the group are called permutations.

Consider the symmetric group of degree 3:

$$S_3 =< a, x \mid a^2 = e, x^3 = e, xa = ax^{-1} >$$

with identity e, generators a and x, and relations $a^2 = e$, $x^3 = e$, and $xa = ax^{-1}$.

This group is also known as the dihedral group of order 6, denoted D_6. Which group of symmetries is it (isomorphic to)?

Answer: This group is isomorphic to the **group of symmetries of an equilateral triangle**. The element x corresponds to a 120° rotation, and a corresponds to a reflection.

□

Fill in the product table below with the given elements. Enter x^2 as "xx" and x^2a as "xxa".

Answer: To fill the table, we use the given relations $a^2 = e$, $x^3 = e$, and $xa = ax^{-1}$. Since $x^3 = e$, we know that $x^{-1} = x^2$. So, the main relation becomes: $xa = ax^2$. From this, we can also derive $x^2a = x(xa) = x(ax^2) = (xa)x^2 = (ax^2)x^2 = ax^4 = ax^3x = aex = ax$. The key relations for simplification are:

- $xa = axx$
- $xxa = ax$

$\circ$	e	x	xx	a	xa	xxa
e	e	x	xx	a	xa	xxa
x	x	xx	e	xa	xxa	a
xx	xx	e	x	xxa	a	xa
a	a	xxa	xa	e	xx	x
xa	xa	a	xxa	x	e	xx
xxa	xxa	xa	a	xx	x	e

□

Select all subgroups of S_3:

A. $\{e, xa\}$

B. $\{e, a, xa, x^2a\}$

C. $\{a, x, xa\}$

D. $\{e, x, xa\}$

E. $\{e, x, x^2, a, xa, x^2a\}$

F. $\{e, x^2, x^2a\}$

G. $\{e, a\}$

H. $\{x\}$

I. $\{e, x^2a\}$

J. $\{x, x^2, a, xa, x^2a\}$

K. $\{e, a, xa\}$

L. $\{e, x, x^2\}$

M. $\{e\}$

N. $\{a, x\}$

Answer: A subset is a subgroup if it contains the identity element e, is closed under the group operation, and contains the inverse of each of its elements. The group S_3 has order 6, so by Lagrange's Theorem, any subgroup must have an order that divides 6 (i.e., order 1, 2, 3, or 6).

- **C, H, J, N**: These are not subgroups because they do not contain the identity element e.
- **B**: This set has 4 elements. 4 does not divide 6, so it cannot be a subgroup.
- **D**: Not closed. $x \circ xa = x^2a$, which is not in the set.
- **F**: Not closed. $x^2 \circ x^2a = xa$, which is not in the set.
- **K**: Not closed. $a \circ xa = x^2$, which is not in the set.

The remaining options are all valid subgroups:

- **A.** $\{e, xa\}$: A subgroup of order 2.
- **E.** $\{e, x, x^2, a, xa, x^2a\}$: The entire group (a subgroup of itself).
- **G.** $\{e, a\}$: A subgroup of order 2.
- **I.** $\{e, x^2a\}$: A subgroup of order 2.
- **L.** $\{e, x, x^2\}$: The subgroup of rotations, order 3.
- **M.** $\{e\}$: The trivial subgroup, order 1.

The correct selections are **A, E, G, I, L, M**.

$\square$

Consider the dihedral group of order 8:

$$D_8 =< a, x \mid a^2 = e, x^4 = e, axa^{-1} = x^{-1} >$$

with identity e, generators a and x, and relations $a^2 = e$, $x^4 = e$, and $axa^{-1} = x^{-1}$.

What is the pattern in the relations for the dihedral groups? Are the differences in the third relation meaningful or not?

Answer: The relations define the dihedral group D_{2n}, which represents the symmetries of a regular n-gon.

- $a^2 = e$ means the generator a (a reflection) has order 2.
- $x^n = e$ means the generator x (a rotation) has order n. In this problem, $n = 4$. For $S_3 \cong D_6$ (symmetries of a triangle), this relation was $x^3 = e$.
- $axa^{-1} = x^{-1}$ is the "interaction" rule. Since $a^{-1} = a$ (from $a^2 = e$), this can be rewritten as $ax = x^{-1}a$.

The pattern is that all dihedral groups D_{2n} share the $a^2 = e$ and $ax = x^{-1}a$ relations. The difference, $x^n = e$, is meaningful as it specifies *which* n-gon the group describes (e.g., $n = 3$ for a triangle, $n = 4$ for a square).

$\square$

Fill in the product table below with the given elements.

Answer: The table is completed using the relations: $x^4 = e$, $a^2 = e$, and $ax^k = x^{-k}a$ (which gives $ax = x^3a$, $ax^2 = x^2a$, $ax^3 = xa$). The inverse relation $x^ka = ax^{-k}$ is also used (e.g., $xa = ax^3$, $x^2a = ax^2$, $x^3a = ax$). The operation is (row) $\circ$ (column).

$\circ$	e	x	x^2	x^3	a	ax	ax^2	ax^3
e	e	x	x^2	x^3	a	ax	ax^2	ax^3
x	x	x^2	x^3	e	ax^3	a	ax	ax^2
x^2	x^2	x^3	e	x	ax^2	ax^3	a	ax
x^3	x^3	e	x	x^2	ax	ax^2	ax^3	a
a	a	ax	ax^2	ax^3	e	x	x^2	x^3
ax	ax	ax^2	ax^3	a	x^3	e	x	x^2
ax^2	ax^2	ax^3	a	ax	x^2	x^3	e	x
ax^3	ax^3	a	ax	ax^2	x	x^2	x^3	e

$\square$

Which group of symmetries is this group (isomorphic to)? What does that tell you about the subgroups? Don't forget the previous two problems!

Answer: This group, D_8, is isomorphic to the **group of symmetries of a square**. The element x represents a 90° rotation, and a represents a reflection. The group has 8 elements: 4 rotations ($\{e, x, x^2, x^3\}$) and 4 reflections ($\{a, ax, ax^2, ax^3\}$).

This tells us that the subgroups will correspond to the symmetries of a square. For example, the set of all rotations ($\{e, x, x^2, x^3\}$) forms a subgroup. The set containing the identity and a 180° rotation ($\{e, x^2\}$) is also a subgroup.

$\square$

Select all subgroups of D_8:

A. $\{x, a\}$

B. $\{e\}$

C. $\{e, a, ax, ax^2, ax^3\}$

D. $\{e, ax^2\}$

E. $\{e, x^2, ax, ax^3\}$

F. $\{e, x, x^2, x^3\}$

G. $\{e, x, x^3, a, ax, ax^3\}$

H. $\{e, ax^3\}$

I. $\{e, ax^2, ax^3\}$

J. $\{e, x^2\}$

K. $\{e, x^2, a, ax^2\}$

L. $\{e, x, x^2, x^3, a, ax, ax^2, ax^3\}$

M. $\{e, a\}$

N. $\{e, x\}$

O. $\{e, ax\}$

P. $\{e, a, ax\}$

Answer: A subset is a subgroup if it contains the identity e, is closed under the operation, and contains all its elements' inverses. By Lagrange's Theorem, a subgroup's order must divide the group's order (which is 8). Thus, we are looking for subgroups of order 1, 2, 4, or 8.

- **Order 1:** $\{e\}$ (the trivial subgroup).
- **Order 2:** $\{e, g\}$ where g has order 2. The elements of order 2 are x^2 (180° rotation) and the four reflections a, ax, ax^2, ax^3.
- **Order 4:** Can be cyclic (like the rotations) or non-cyclic (like V_4).
- **Order 8:** The group itself.

The correct subgroups are:

- **B.** $\{e\}$: The trivial subgroup (order 1).
- **D.** $\{e, ax^2\}$: Subgroup of order 2 (identity + a reflection).
- **E.** $\{e, x^2, ax, ax^3\}$: Subgroup of order 4 (identity, 180° rotation, and the two diagonal reflections).
- **F.** $\{e, x, x^2, x^3\}$: The cyclic subgroup of all rotations (order 4).
- **H.** $\{e, ax^3\}$: Subgroup of order 2 (identity + a reflection).
- **J.** $\{e, x^2\}$: Subgroup of order 2 (identity + 180° rotation).
- **K.** $\{e, x^2, a, ax^2\}$: Subgroup of order 4 (identity, 180° rotation, and the horizontal/vertical reflections).
- **L.** $\{e, x, x^2, x^3, a, ax, ax^2, ax^3\}$: The entire group (order 8).
- **M.** $\{e, a\}$: Subgroup of order 2 (identity + a reflection).
- **O.** $\{e, ax\}$: Subgroup of order 2 (identity + a reflection).

(Options A, P fail as they lack the identity. C, G fail as their order does not divide 8. I, N, P fail the closure property.)

□

Let G be a group.

Let $a \in G$. We can prove that the set $A = \{a^i \mid i \in \mathbb{Z}\}$ is a subgroup of G called the **cyclic subgroup generated by** a, denoted $\langle a \rangle$.

Definition: The group G is called **cyclic** if (and only if) there exists $a \in G$ such that every $x \in G$ is a power of a. If G is cyclic and $G = \langle a \rangle$, then a is a **generator** of G.

What is the order of each of the elements in the following groups? (See previous problems for the group presentations in generators and relations.)

V_4, the Klein 4-group:

1. What is the order of a?
2. What is the order of e?
3. What is the order of ab?
4. What is the order of b?

Options:

A. 4

B. 3

C. 1

D. 2

E. None of the above

Answer: The order of an element g is the smallest positive integer m such that $g^m = e$. The Klein 4-group $V_4 = \{e, a, b, ab\}$ is defined by the relations $a^2 = e$, $b^2 = e$, and $(ab)^2 = e$.

1. **Order of a:** The relation is $a^2 = e$. Since $a^1 = a \neq e$, the smallest such integer is 2. The correct option is **D**.
2. **Order of e:** The identity element e always has order 1, since $e^1 = e$. The correct option is **C**.
3. **Order of ab:** The relation is $(ab)^2 = e$. Since $(ab)^1 = ab \neq e$, the smallest such integer is 2. The correct option is **D**.
4. **Order of b:** The relation is $b^2 = e$. Since $b^1 = b \neq e$, the smallest such integer is 2. The correct option is **D**.

(Note: The answers selected in the image, E, A, E, B, are incorrect.)

□

If V_4 is cyclic, select all elements which generate the group.

A. b

B. V_4 is not cyclic.

C. e

D. a

E. ab

Answer: A group is cyclic if there exists an element (a generator) whose order is equal to the order of the group. The group V_4 has order 4. Therefore, to be cyclic, it must have an element of order 4.

From the previous problem, the orders of the elements in $V_4 = \{e, a, b, ab\}$ are:

- $o(e) = 1$
- $o(a) = 2$

- $o(b) = 2$
- $o(ab) = 2$

Since V_4 has no element of order 4, no single element can generate the entire group.

The correct selection is **B.** V_4 **is not cyclic.**

$\square$

S_3, the symmetric group of degree 3:

What is the order of the following elements?

1. x^2
2. xa
3. e
4. x
5. x^2a
6. a

Options:

A. 1

B. 2

C. 3

D. 4

E. 6

F. 5

G. None of the above

Answer: The order of an element g is the smallest positive integer m such that $g^m = e$. From the previous problem, S_3 is given by the relations $a^2 = e$, $x^3 = e$, and $xa = ax^{-1} = ax^2$. We also derived $x^2a = ax$.

1. **Order of x^2:**
   - $(x^2)^1 = x^2 \neq e$
   - $(x^2)^2 = x^4 = x^3 \cdot x = e \cdot x = x \neq e$
   - $(x^2)^3 = x^6 = (x^3)^2 = e^2 = e$

   The order is **3**. (Option **C**)
2. **Order of xa:**
   - $(xa)^1 = xa \neq e$
   - $(xa)^2 = (xa)(xa) = x(ax)a = x(x^2a)a = x^3a^2 = (e)(e) = e$

   The order is **2**. (Option **B**)
3. **Order of e:**
   - $e^1 = e$

   The order is **1**. (Option **A**)
4. **Order of x:**
   - $x^1 = x \neq e$
   - $x^2 = x^2 \neq e$
   - $x^3 = e$ (by relation)

   The order is **3**. (Option **C**)
5. **Order of x^2a:**

- $(x^2a)^1 = x^2a \neq e$
- $(x^2a)^2 = (x^2a)(x^2a) = x^2(ax^2)a = x^2(xa)a = x^3a^2 = (e)(e) = e$

The order is **2**. (Option **B**)

6. **Order of a:**
   - $a^1 = a \neq e$
   - $a^2 = e$ (by relation)

   The order is **2**. (Option **B**)

□

If S_3 is cyclic, select all elements which generate the group.

A. x^2a

B. S_3 is not cyclic.

C. a

D. x

E. e

F. xa

G. x^2

Answer: The group S_3 has order 6. To be cyclic, it must have an element of order 6.

From the previous problem, the orders of the elements in $S_3 = \{e, x, x^2, a, xa, x^2a\}$ are:

- $o(e) = 1$
- $o(x) = 3$
- $o(x^2) = 3$
- $o(a) = 2$
- $o(xa) = 2$
- $o(x^2a) = 2$

Since S_3 has no element of order 6, no single element can generate the entire group. (Note also that S_3 is non-abelian, and all cyclic groups are abelian, which is another proof.)

The correct selection is **B. S_3 is not cyclic.**

□

D_8, the dihedral group of order 8:

What is the order of the following elements?

1. a
2. ax^2
3. ax
4. ax^3
5. e
6. x^2
7. x
8. x^3

Options:

A. 8

B. 7

C. 6

D. 5

E. 4

F. 3

G. 1

H. 2

I. None of the above

Answer: The order of an element g is the smallest positive integer m such that $g^m = e$. From the previous problem, D_8 is given by the relations $a^2 = e$, $x^4 = e$, and $ax = x^{-1}a = x^3a$. We also use the derived relations $x^k a = ax^{-k}$ and $(ax^k)^2 = e$ for any k.

1. **Order of a:**
   - $a^1 = a \neq e$
   - $a^2 = e$ (by relation)

   The order is **2**. (Option **H**)
2. **Order of ax^2:**
   - $(ax^2)^1 = ax^2 \neq e$
   - $(ax^2)^2 = (ax^2)(ax^2) = a(x^2a)x^2 = a(ax^{-2})x^2 = a(ax^2)x^2 = a^2x^4 = (e)(e) = e$

   The order is **2**. (Option **H**)
3. **Order of ax:**
   - $(ax)^1 = ax \neq e$
   - $(ax)^2 = (ax)(ax) = a(xa)x = a(ax^{-1})x = a(ax^3)x = a^2x^4 = (e)(e) = e$

   The order is **2**. (Option **H**)
4. **Order of ax^3:**

- $(ax^3)^1 = ax^3 \neq e$
- $(ax^3)^2 = (ax^3)(ax^3) = a(x^3a)x^3 = a(ax^{-3})x^3 = a(ax)x^3 = a^2x^4 = (e)(e) = e$

The order is **2**. (Option **H**)

5. **Order of e:**
   - $e^1 = e$

   The order is **1**. (Option **G**)

6. **Order of x^2:**
   - $(x^2)^1 = x^2 \neq e$
   - $(x^2)^2 = x^4 = e$ (by relation)

   The order is **2**. (Option **H**)

7. **Order of x:**
   - $x^1 = x \neq e$
   - $x^2 = x^2 \neq e$
   - $x^3 = x^3 \neq e$
   - $x^4 = e$ (by relation)

   The order is **4**. (Option **E**)

8. **Order of x^3:**
   - $(x^3)^1 = x^3 \neq e$
   - $(x^3)^2 = x^6 = x^4 \cdot x^2 = e \cdot x^2 = x^2 \neq e$
   - $(x^3)^3 = x^9 = (x^4)^2 \cdot x = e^2 \cdot x = x \neq e$
   - $(x^3)^4 = x^{12} = (x^4)^3 = e^3 = e$

   The order is **4**. (Option **E**)

□

If D_8 is cyclic, select all elements which generate the group.

A. e

B. ax^3

C. ax

D. D_8 is not cyclic.

E. a

F. x^3

G. x^2

H. ax^2

I. x

Answer: A group is cyclic if there is an element (a generator) whose order is equal to the order of the group. The group D_8 has order 8. For it to be cyclic, it must have an element of order 8.

From the previous problem, the orders of the elements in D_8 are:

- $o(e) = 1$
- $o(x) = 4$
- $o(x^2) = 2$
- $o(x^3) = 4$
- $o(a) = 2$
- $o(ax) = 2$
- $o(ax^2) = 2$
- $o(ax^3) = 2$

Since D_8 has no element of order 8, no single element can generate the entire group. (Additionally, D_8 is non-abelian, and all cyclic groups are abelian.)

The correct selection is **D. D_8 is not cyclic.**

$\square$

$\mathbb{Z}_5$, under addition:

What is the order of the following elements?

1. $[1]$
2. $[2]$
3. $[4]$
4. $[3]$
5. $[0]$

Options:

A. 4

B. 3

C. 1

D. 5

E. 2

F. None of the above

Answer: The order of an element g in an additive group is the smallest positive integer m such that $m \cdot g = e$, where e is the identity element. For $(\mathbb{Z}_5, +)$, the identity e is $[0]$.

Since 5 is a prime number, $(\mathbb{Z}_5, +)$ is a cyclic group of order 5. By a property of cyclic groups (or Lagrange's Theorem), the order of any non-identity element must divide the group order, 5. Since 5 is prime, the only possible orders are 1 or 5.

1. **Order of $[1]$:** $[1]$ is not the identity. $5 \cdot [1] = [1] + [1] + [1] + [1] + [1] = [5] = [0]$. The order is **5**. (Option **D**)
2. **Order of $[2]$:** $[2]$ is not the identity. $5 \cdot [2] = [2] + [2] + [2] + [2] + [2] = [10] = [0]$. The order is **5**. (Option **D**)
3. **Order of $[4]$:** $[4]$ is not the identity. $5 \cdot [4] = [4] + [4] + [4] + [4] + [4] = [20] = [0]$. The order is **5**. (Option **D**)
4. **Order of $[3]$:** $[3]$ is not the identity. $5 \cdot [3] = [3] + [3] + [3] + [3] + [3] = [15] = [0]$. The order is **5**. (Option **D**)
5. **Order of $[0]$:** $[0]$ is the identity element. $1 \cdot [0] = [0]$. The order is **1**. (Option **C**)

□

If the group $\mathbb{Z}_5$ under addition is cyclic, select all elements which generate the group.

A. $\mathbb{Z}_5$ under addition is not cyclic.

B. $[3]$

C. $[1]$

D. $[0]$

E. $[4]$

F. $[2]$

Answer: The group $(\mathbb{Z}_5, +)$ is a cyclic group of order 5. (Option A is false). The generators of a cyclic group $(\mathbb{Z}_n, +)$ are all elements $[k]$ such that $\gcd(k, n) = 1$. Here $n = 5$. We need to find all $k \in \{0, 1, 2, 3, 4\}$ such that $\gcd(k, 5) = 1$.

- $\gcd(0, 5) = 5 \neq 1$. (Not a generator).
- $\gcd(1, 5) = 1$. (Generator).
- $\gcd(2, 5) = 1$. (Generator).

- $\gcd(3, 5) = 1$. (Generator).
- $\gcd(4, 5) = 1$. (Generator).

The generators are [1], [2], [3], and [4].

The correct selections are **B. [3], C. [1], E. [4], and F. [2]**.

$\square$

$\mathbb{Z}_6$, under addition:

What is the order of the following elements?

1. [5]
2. [1]
3. [2]
4. [0]
5. [3]
6. [4]

Options:

A. 2

B. 4

C. 3

D. 5

E. 1

F. 6

G. None of the above

Answer: The order of an element $[k]$ in $(\mathbb{Z}_n, +)$ is the smallest positive integer m such that $m \cdot [k] = [0]$. The identity element e is $[0]$. The order of $[k]$ can be quickly calculated with the formula:

$$\text{order}([k]) = \frac{n}{\gcd(n,k)}$$

Here, $n = 6$.

1. **Order of $[5]$:** $6/\gcd(6,5) = 6/1 = 6$. The order is **6**. (Option **F**)
2. **Order of $[1]$:** $6/\gcd(6,1) = 6/1 = 6$. The order is **6**. (Option **F**)
3. **Order of $[2]$:** $6/\gcd(6,2) = 6/2 = 3$. The order is **3**. (Option **C**)
4. **Order of $[0]$:** $[0]$ is the identity element. The order of the identity is always **1**. (Option **E**)
5. **Order of $[3]$:** $6/\gcd(6,3) = 6/3 = 2$. The order is **2**. (Option **A**)
6. **Order of $[4]$:** $6/\gcd(6,4) = 6/2 = 3$. The order is **3**. (Option **C**)

□

If the group $\mathbb{Z}_6$ under addition is cyclic, select all elements which generate the group.

A. [2]

B. $\mathbb{Z}_6$ under addition is not cyclic.

C. [5]

D. [3]

E. [0]

F. [4]

G. [1]

Answer: The group $(\mathbb{Z}_6, +)$ is a cyclic group of order 6. (Option B is false). The generators of a cyclic group $(\mathbb{Z}_n, +)$ are all elements $[k]$ such that $\gcd(k, n) = 1$. Here $n = 6$. We need to find all $k \in \{0, 1, 2, 3, 4, 5\}$ such that $\gcd(k, 6) = 1$.

- $\gcd(0, 6) = 6 \neq 1$.
- $\gcd(1, 6) = 1$. (Generator).
- $\gcd(2, 6) = 2 \neq 1$.
- $\gcd(3, 6) = 3 \neq 1$.
- $\gcd(4, 6) = 2 \neq 1$.
- $\gcd(5, 6) = 1$. (Generator).

The generators are [1] and [5].

The correct selections are **C. [5] and G. [1]**.

$\square$

$\mathbb{Z}_5^*$, under multiplication:

What is the order of the following elements?

1. [4]
2. [1]
3. [2]
4. [3]

Options:

A. 5

B. 2

C. 3

D. 1

E. 4

F. None of the above

Answer: The group is $(\mathbb{Z}_5^*, \cdot)$. Since 5 is prime, the elements are $\mathbb{Z}_5^* = \{[1], [2], [3], [4]\}$. The operation is multiplication modulo 5. The identity element e is [1]. The order of an element g is the smallest positive integer m such that $g^m = [1]$.

1. **Order of [4]:**
   - $[4]^1 = [4] \neq [1]$
   - $[4]^2 = [16] = [1]$

   The order is **2**. (Option **B**)
2. **Order of [1]:**
   - [1] is the identity element.

   The order is **1**. (Option **D**)
3. **Order of [2]:**
   - $[2]^1 = [2] \neq [1]$
   - $[2]^2 = [4] \neq [1]$
   - $[2]^3 = [8] = [3] \neq [1]$
   - $[2]^4 = [16] = [1]$

   The order is **4**. (Option **E**)
4. **Order of [3]:**
   - $[3]^1 = [3] \neq [1]$
   - $[3]^2 = [9] = [4] \neq [1]$
   - $[3]^3 = [27] = [2] \neq [1]$
   - $[3]^4 = [81] = [1]$

   The order is **4**. (Option **E**)

(The selections shown in the image, B, D, E, E, are all correct.)

□

If the group $\mathbb{Z}_5^*$ under multiplication is cyclic, select all elements which generate the group.

A. [4]

B. $\mathbb{Z}_5^*$ under multiplication is not cyclic.

C. [3]

D. [1]

E. [2]

Answer: The group $(\mathbb{Z}_5^*, \cdot) = \{[1], [2], [3], [4]\}$ has order 4. The group $\mathbb{Z}_p^*$ (for p prime) is always cyclic. (Option B is false). We need to find the generators, which are the elements of order 4.

- **Order of [1]:** This is the identity. Order is 1.
- **Order of [2]:** $[2]^1 = [2]$, $[2]^2 = [4]$, $[2]^3 = [8] = [3]$, $[2]^4 = [16] = [1]$. Order is 4. (Generator).
- **Order of [3]:** $[3]^1 = [3]$, $[3]^2 = [9] = [4]$, $[3]^3 = [12] = [2]$, $[3]^4 = [6] = [1]$. Order is 4. (Generator).
- **Order of [4]:** $[4]^1 = [4]$, $[4]^2 = [16] = [1]$. Order is 2.

The generators are [2] and [3].

The correct selections are **C. [3] and E. [2]**.

□